

# OVERVIEW OF RESEARCH

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# Steps to conducting research

1. Read relevant literature, to get necessary background and to learn what else has been done
2. Identify a concrete investigative question
3. Identify an appropriate methodology for answering the question
4. Carry out the actual research project
5. Disseminate results through oral and written communication

## Example #1:

1. Question: Does MySQL or MongoDB enable more efficient retrieval of genomic data (or football data, or social media data, ...)?
2. Methodology: Load the same data into both databases, use appropriate indexing, and measure performance using relevant queries.

## Example #2:

1. Question: Is GD-sort better than quicksort? (Note: better would have to be defined; e.g., better = faster on random data)
2. Methodology:
  - Theoretical running time and memory, big O notation
  - Comparison of GD-sort and quicksort using real data

# How to read scientific research

- See **Tips for Reading Journal** articles, on page 3 of <https://web.archive.org/web/20230129170522/www.colby.edu/biology/bi319/GuideReadJour.doc>
- Example:
  - **Empirical Performance of Internal Sorting Algorithm**
  - Silas, F., Musa, Y., & Joyce, S. A. (2017). *British Journal of Mathematics and Computer Science*, 20(1), 1-9.
    - [https://www.researchgate.net/publication/312168023\\_Empirical\\_Performance\\_of\\_Internal\\_Sorting\\_Algorithm](https://www.researchgate.net/publication/312168023_Empirical_Performance_of_Internal_Sorting_Algorithm)
  - What does the abstract tell us?
  - What does Figure 1 tell us?
  - What does Figure 2 tell us?
  - What does Figure 4 tell us?

# Computer Science is a broad discipline

- Association for Computing Machinery (ACM) knowledge areas
  - Artificial Intelligence
  - Algorithms
  - Architecture and Organization
  - Graphics and Visualization
  - Human Computer Interaction
  - Programming Languages
  - Networking and Communication
  - Operating Systems
  - Distributed Computing
  - Data Management
  - Software Engineering
  - Software Development
  - Security
  - Ethics

For full list see: <https://csed.acm.org/wp-content/uploads/2024/04/3.1-Body-of-Knowledge-1.pdf>

# Choosing a Research Topic

1. What aspects of computer science interest you?
2. Explore what others have done by reading the literature
  - Media and review articles can be a good place to start, but you will need to find *research* articles
3. Trending topics
  - Chat-GPT
  - Facial recognition
  - AI and ethics
  - Self-driving cars (image recognition)
  - Virtual reality
  - Quantum computing
  - Targeted advertising
  - Blockchain / cryptocurrency

# Computer Science

- Computer Science is the systematic study of the **feasibility, structure, expression**, and mechanization of the methodical processes (or **algorithms**) that underlie the **acquisition, representation, processing, storage, communication of, and access to information...**
- Computer Science is not *just* about building computers or writing computer programs!...**Computer science is** not about the tools we use to carry out computation. It is about **how we use such tools, and what we find out when we do.**
- Source: <http://www.cs.bu.edu/AboutCS/WhatIsCS.pdf> (no longer active)

Computer science has applications in:

Computer Engineering

Information Technology and Information Systems

Bioinformatics and Computational Biology

Computational Statistics

Mathematical Modeling

# What is a computer (part 1)?

- A computer is a *platform* that implements varying algorithms and methodologies for storing, retrieving, and analyzing information
  - Ex: Quicksort is a sorting method that exists and can be analyzed independently of any computer.
  - Ex: The properties of a relational database do not depend on the computer used to store the data
- There is a theoretical component to computer science, but these theories can be tested in practice
- Project ideas:
  - Develop a new method and compare with existing methods (difficult)
  - Compare two methods
  - Evaluate how altering a method changes the behavior / performance of the method

# What is a computer (part 2)?

- A computer is a *tool* that can be used to
  - Analyze (large amounts of) data
    - Example: analyzing IMDB data: <https://www.r-bloggers.com/imdb-movie-analysis/>
    - Example: Can Twitter be used to predict crimes? (Answer – yes, to some extent: <http://www.citylab.com/tech/2014/03/how-twitter-could-help-police-departments-predict-crime/8651/>)
  - Simulate physical or biological system and analyze it
    - Cellular automata models
      - <http://mathworld.wolfram.com/CellularAutomaton.html>
      - A cellular automata model of traffic flow
        - [http://www.ajuronline.org/uploads/Volume%2012/Issue\\_1/AJURVol12Issue1Aug2014pp39to48.pdf](http://www.ajuronline.org/uploads/Volume%2012/Issue_1/AJURVol12Issue1Aug2014pp39to48.pdf)
    - Agent-based models
      - Agents move and interact according to rules executed at discrete time steps
      - In some cases this approach has identified emergent behavior that may be counter-intuitive. Ex: should we put a pillar by the front door of the classroom?  
Ex: "The Game":  
<http://web.archive.org/web/20230208205919/http://www.icosystem.com/labsdemos/the-game/>
      - Write a program to download and analyze a large dataset, such as tweets, to answer a specific question
  - Develop a simulation (or modify an existing one) to answer a specific question



# Where to find published research

- Library Databases: <http://easternct.libguides.com/az.php?s=126319>  
(if off-campus, log in with your Eastern userID and password)
  - ABI Inform Complete: The database features thousands of full-text journals, dissertations, working papers, etc. Includes IEEE journals.
  - ACM digital library: <http://dl.acm.org>
  - ACM surveys (CSUR) <http://dl.acm.org/citation.cfm?id=J204>
    - Provides comprehensive tutorials and survey papers
  - Academic Search Premier: Multidisciplinary database with full text articles
- Undergraduate research:
  - American Journal of Undergrad Research:
    - <http://www.ajuronline.org>
  - Student Pulse
    - <http://www.studentpulse.com/topics/15/computer-science>
- Google Scholar (<http://scholar.google.com>)